

Problem

Given the transmission parameters

$$[\mathbf{T}] = \begin{bmatrix} 3 & 20 \\ 1 & 7 \end{bmatrix}$$

obtain the other five two-port parameters.

Step-by-step solution

Step 1 of 6

We have

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} 3 & 20 \\ 1 & 7 \end{bmatrix}$$

$$\Delta T = AD - CB = 1$$

Step 2 of 6

$$(a) [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} \frac{A}{C} & \frac{\Delta T}{CV} \\ \frac{1}{C} & \frac{D}{C} \end{bmatrix} = \begin{bmatrix} \frac{3}{1} & \frac{1}{1} \\ \frac{1}{1} & \frac{7}{1} \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 7 \end{bmatrix} \Omega$$

Step 3 of 6

$$(b) [Y] = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} = \begin{bmatrix} \frac{D}{B} & \frac{-AT}{B} \\ \frac{-1}{B} & \frac{A}{B} \end{bmatrix} = \begin{bmatrix} \frac{7}{20} & \frac{-1}{20} \\ \frac{-1}{20} & \frac{3}{20} \end{bmatrix} = \begin{bmatrix} \frac{7}{20} & \frac{-1}{20} \\ \frac{-1}{20} & \frac{3}{20} \end{bmatrix} S$$

Step 4 of 6

$$(c) [t] = \begin{bmatrix} t_{11} & t_{12} \\ t_{21} & t_{22} \end{bmatrix} = \begin{bmatrix} \frac{D}{\Delta T} & \frac{B}{\Delta T} \\ \frac{C}{\Delta T} & \frac{A}{\Delta T} \end{bmatrix} = \begin{bmatrix} \frac{7}{1} & \frac{20}{1} \\ \frac{1}{1} & \frac{3}{1} \end{bmatrix} = \begin{bmatrix} 7 & 20 \\ 1S & 3 \end{bmatrix}$$

Step 5 of 6

$$(d) [h] = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \begin{bmatrix} \frac{B}{D} & \frac{\Delta T}{D} \\ \frac{-1}{D} & \frac{C}{D} \end{bmatrix} = \begin{bmatrix} \frac{20}{7} & \frac{1}{7} \\ \frac{-1}{7} & \frac{1}{7} \end{bmatrix} = \begin{bmatrix} \frac{20}{7} \Omega & \frac{1}{7} \\ \frac{-1}{7} & \frac{1}{7S} \end{bmatrix}$$

Step 6 of 6

$$(e) \quad [g] = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} = \begin{bmatrix} \frac{C}{A} & \frac{\Delta T}{A} \\ \frac{1}{A} & \frac{B}{A} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} & \frac{-1}{3} \\ \frac{1}{3} & \frac{20}{3} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} S & \frac{-1}{3} \\ \frac{1}{3} & \frac{20}{3} \Omega \end{bmatrix}$$